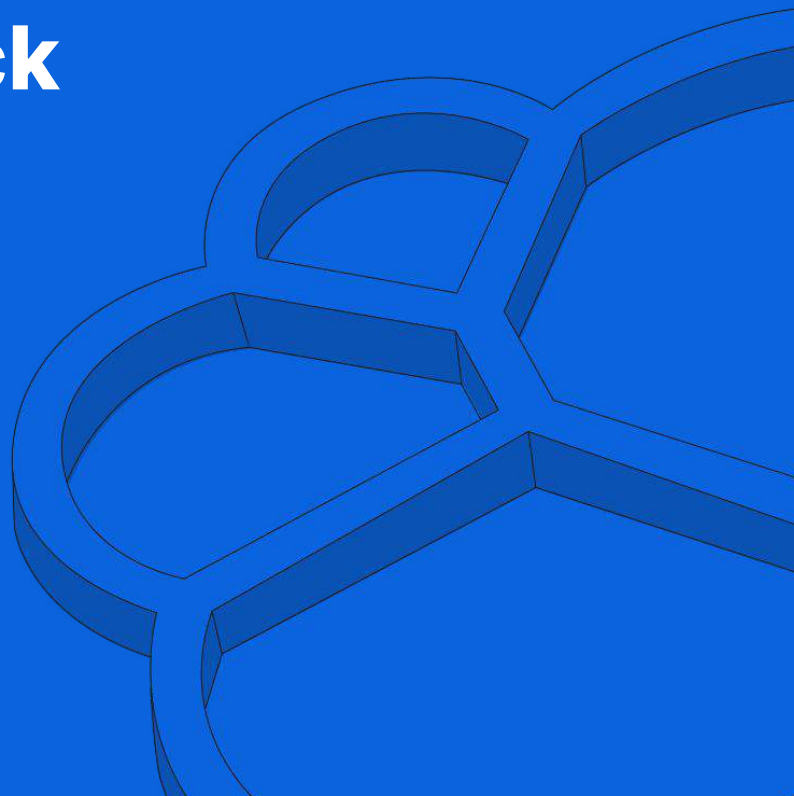
Multimodal Ingest: Apache Tika 4.x in Your Search Stack

Elastic Washington DC User Group

November 10, 2026

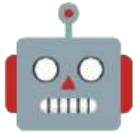
Tim Allison

elastic | The Search AI Company



What's Changed In File Processing

- LLMs/RAG/Agentic search
- Multimodal embeddings
- VLMs
- More people care → abundance of tools
- A new user in the room →

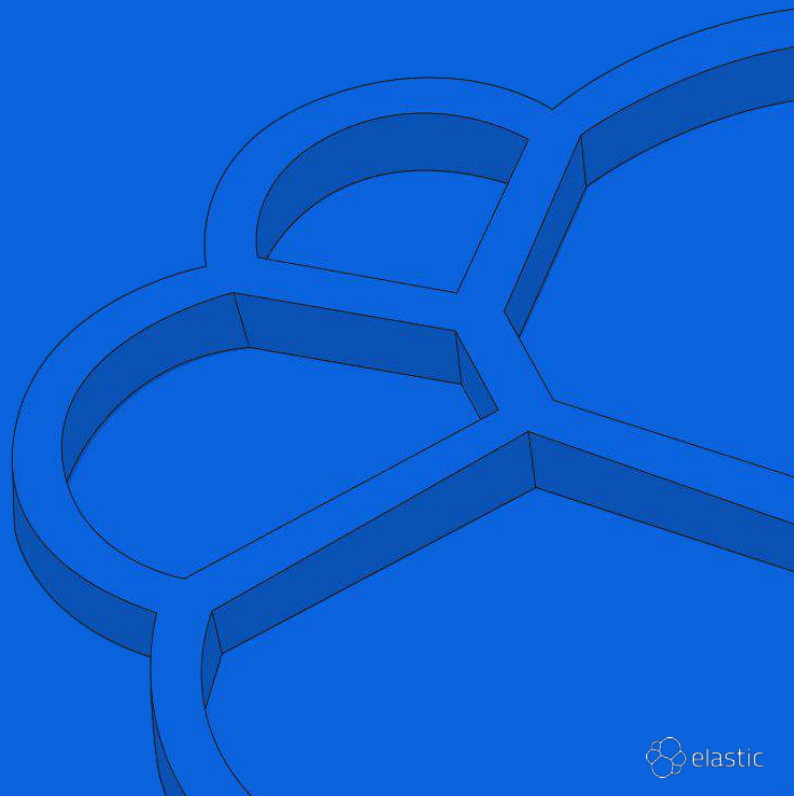


What Hasn't Changed

- Content extraction quality matters
- Scaling, reliability, integration
- Recursively handling embedded files

What's new in Tika 4.x?





Pushing Intelligence into Parsers/Detectors



Pushing intelligence into the parsers

- Text/binary classifier in tika-core
- Junk – if you see something, fix something… *during* the parse
- Improving PDF structural extraction (tbd)
- See also final section on vlms etc…

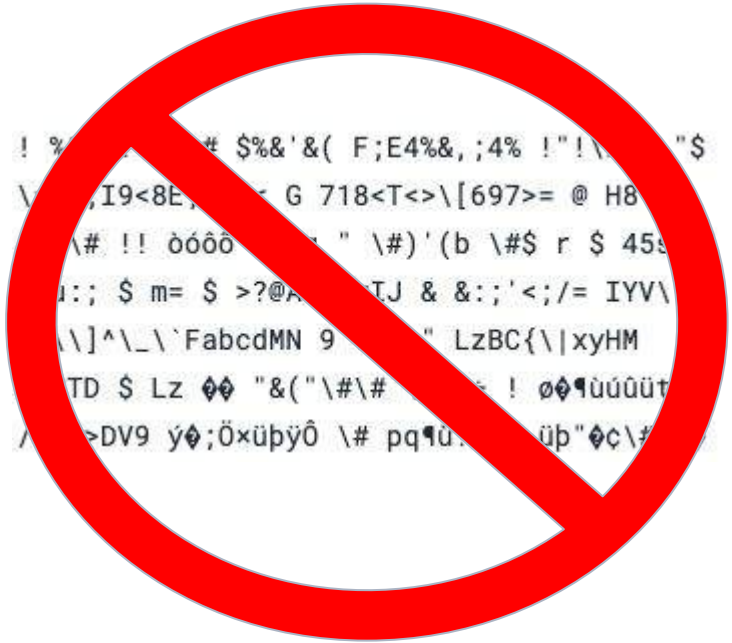
Junk Remediation: Charset Detection

Use junk detector to select header vs statistical encoding

Source	Detected	Decoded String
Meta header	<meta charset= "windows-1251">	P  PSC ‚PµCЪPSPµC‚-PjP° PiP°P·PëPS
Statistical/ Universal chardet	UTF-8	Интернет-магазин

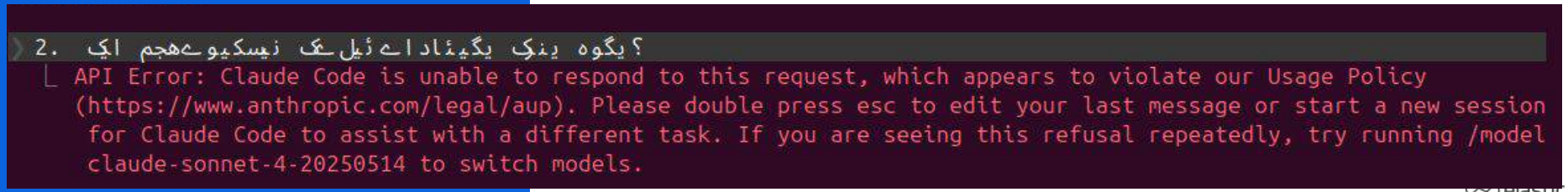
Junk remediation: PDFs, when to trigger OCR (planned 4.3?)

- 3.x: character count < x || % of missing unicode mappings > y
- 4.3+: ...and junk detector



Digression: Junk and agents





Claude review of these slides 🤣



Issues to address

1. **The Urdu/Claude Code slide (slide 21, "Junk and agents")**: This is your riskiest slide. It shows an Anthropic API usage policy error mid-translation. The audience will have questions: *why did it refuse? is Urdu text flagged as a safety issue?* This could derail you into a side conversation about AI safety policy that has nothing to do with Tika. Either replace the example with something less sensitive, or add explicit narration about what happened and why it's relevant.

Leaning into safety and scaling with tika-pipes



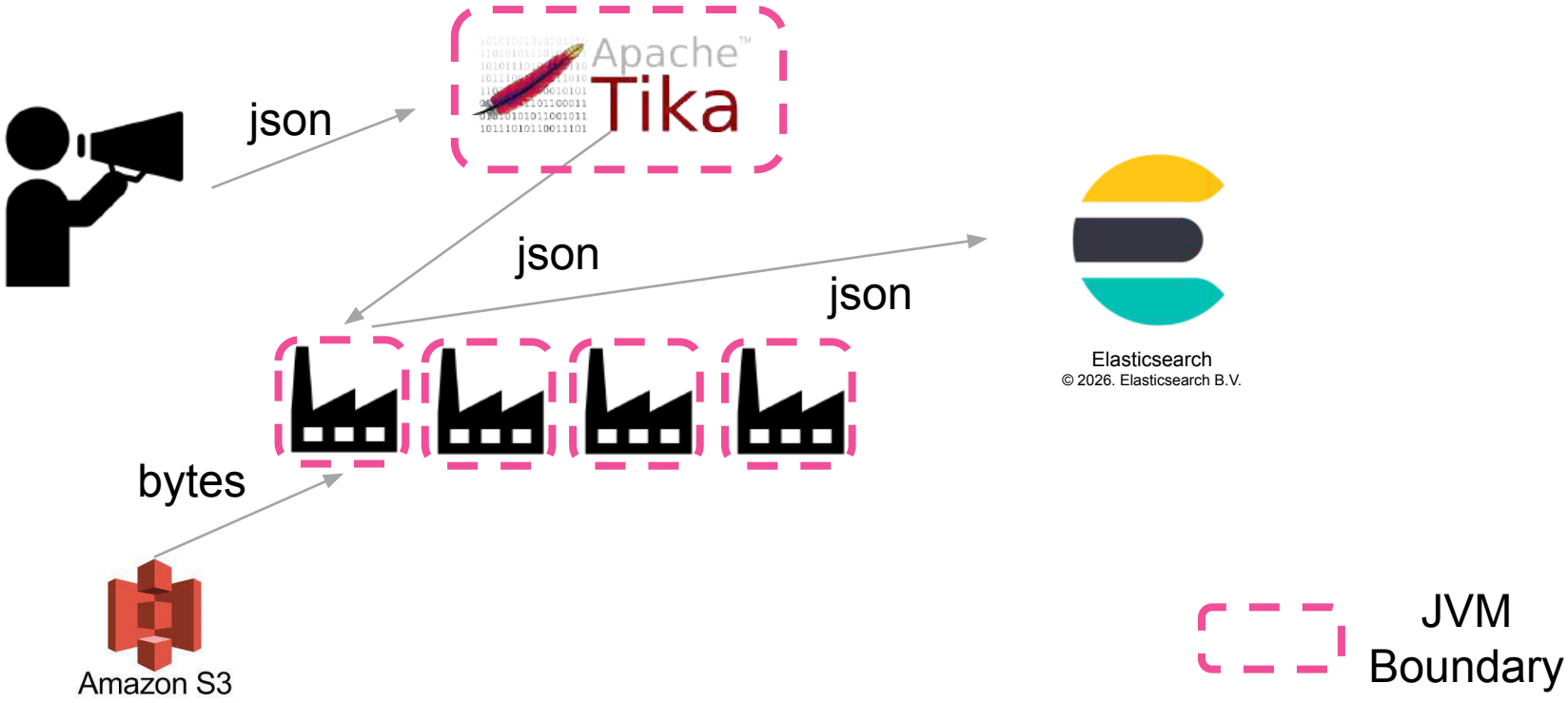
Forked parsing in 3.x

- tika-batch
- ForkParser
- tika-server
- tika-pipes

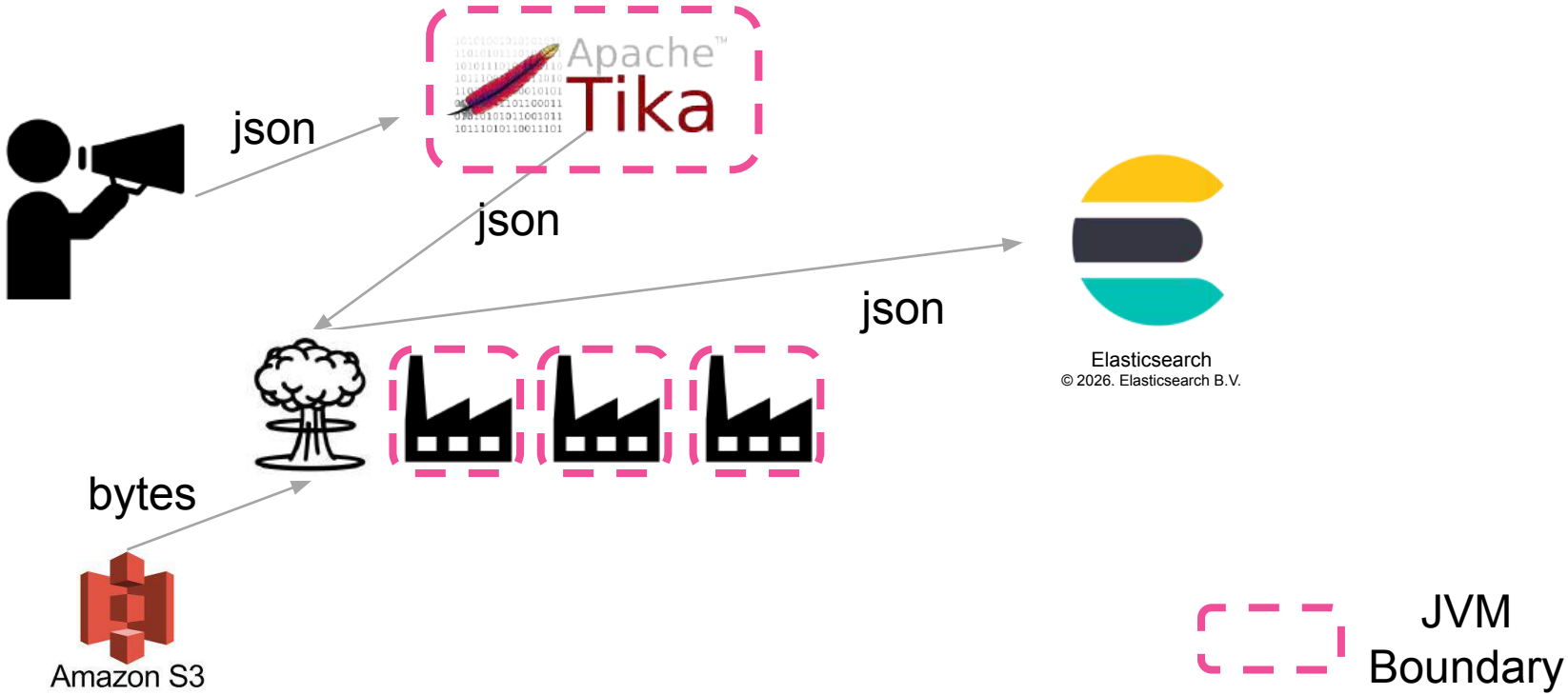
Forked parsing in 4.x

- ~~tika-batch~~
- ~~ForkParser~~
- ~~tika-server~~
- tika-pipes

Fetchers/emitters: Process isolation per parse

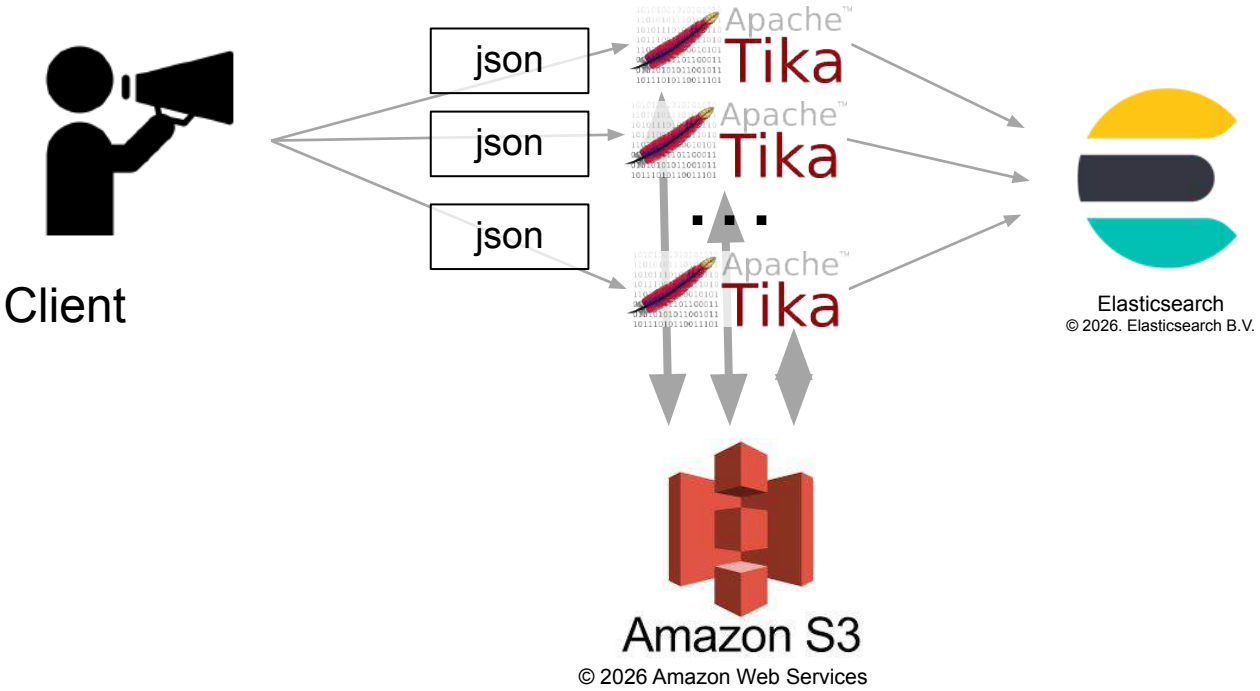


Fetchers/emitters: Process isolation per parse



© 2026 Amazon Web Services

Fetchers/emitters: Scaling across a cluster



© 2026. Elasticsearch B.V.
© 2026 Amazon Web Services

Define an S3 fetcher

```
{
  "fetchers": {
    "s3f": {
      "s3-fetcher": {
        "bucket": "my-tika-input",
        "region": "us-east-1",
        "prefix": "incoming/",
        "credentialsProvider":
          "instance",
        "extractUserMetadata": true,
        "spoolToTemp": true
      }}}}
```

Define an Elasticsearch emitter

```
{
  "emitters": {
    "ese": {
      "es-emitter": {
        "esUrl": "[link removed]",
        "idField": "doc_id",
        "attachmentStrategy":
          "SEPARATE_DOCUMENTS",
        "updateStrategy": "OVERWRITE",
        "chunkStrategy": "INLINE",
        "apiKey": "${env:ES_API_KEY}",
        "httpClientConfig": {
        "connectionTimeoutMillis":
          10000,
        "socketTimeoutMillis": 60000
}}}}}
```

Example FetchEmitTuple

```
{
  "id": "q3",
  "fetcher": "s3f",
  "fetchKey": "reports/q3.pdf",
  "emitter": "ese",
  "emitKey": "q3"
}
```

With the prefix “incoming/” in the s3 config, read this file from s3:

```
s3://my-tika-input/incoming/reports/q3.pdf
```

Tika pipes everywhere

- PipesForkParser
- tika-app
- tika-server (REST): /pipes /async /unpack and even /rmeta and /tika

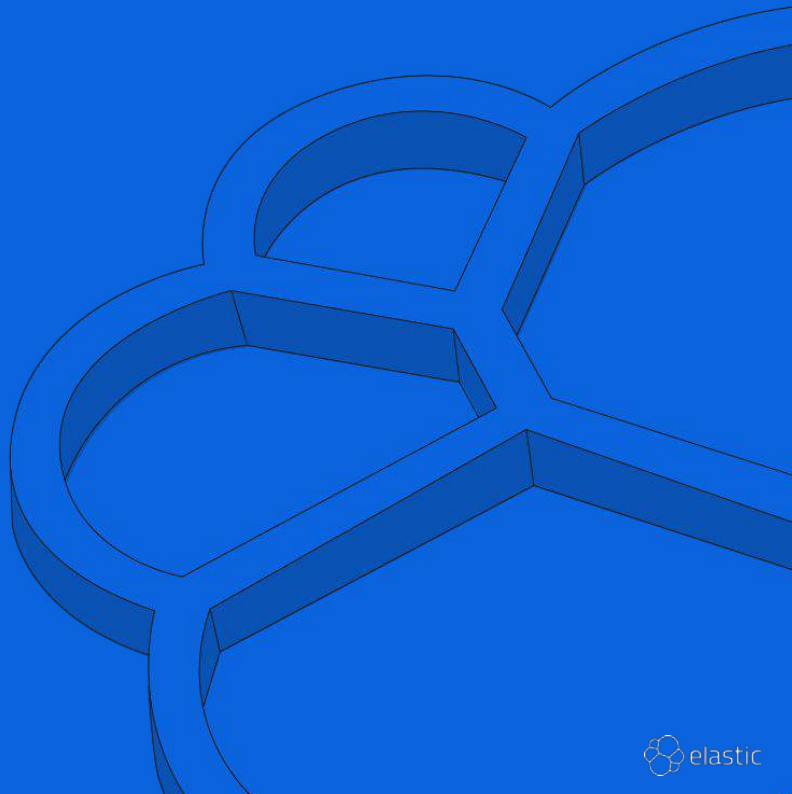
And with Pipes everywhere in tika-server

- Backpressure signal: 429
- And early warning in Prometheus metrics (Micrometer): share of pipes workers busy

Tika - gRPC via Nicholas DiPiazza

- Alternative to REST tika-server
- tika-pipes native
- Container and k8s ready
- Distributed configuration via Ignite
- Hot-pluggable fetchers and emitters
- MCP-ready

Integrations with OCR/VLM APIs



First, VLMs vs ColPali-like late interaction models

- VLMs at index time vs search time
- ColPali/ColQwen – multiple vectors per page and one vector per query token → summed maxsim (late interaction)
- Original ColPali has ~1k vectors per page/image

VLMs at index time: general vs specialized

- Rebuild the cow from the hamburger – turning PDF word soup into logical structure
- Label/describe images
- Extract form fields
- Extract numbers from charts
- OCR

Tika's Legacy OCR hook for tesseract

- Image parser looks for OCR parser and applies it to every image no matter how deeply embedded zip→msg→jpeg
- PDFs
  - Four modes: AUTO, EXTRACT, OCR, EXTRACT_AND_OCR, NONE
  - AUTO: Per page, extract text, if not enough text or missing unicode mappings, render page and then run OCR

Missing Unicode Mappings

PDF

JUDICIAL federalism relies on the principle that the state and federal courts together comprise an integrated system for the delivery of justice in the United States. Historically, the two court systems have played different but equally significant roles in our federal system. The state courts have served as the primary forums for resolving civil disputes and the chief tribunals for enforcing the criminal law. The federal

Text Copied from Adobe Reader
(Very similar to Apache Tika's text)

```
7KH·-
XGLFLDO·&RQIHUHQFH·RI·WKH·8QLWHG·6WDWHV·KDV·DSSURY
HG·WKH·UHFRPPHQGDWLRQV·DQG·LPSOHPHQWDWLRQ·VWUDWHJL
HV·LQ·WKLV·/RJ·5DQJH·3ODQ·WR·JXLGH·IXWXUH
DGPLQLVWUDWLYH·DFWLRQ·DQG·SROLF\·GHYHORSPHQW·E
\·WKH·&RQIHUHQFH·DQG·RWKHU·MXGLFLDO·EUDQFK·DXWKRUL
WLHV···$OO·RWKHU·WH
[W·LQ·WKLV·3ODQ··LQFOXGLQJ·FRPPHQWDU\·RQ
LQGLYLGXDO·UHFRPPHQGDWLRQV·DQG·VWUDWHJLHV··H
[SODLQV·DQG·VXSSOHPHQW·WKH·DSSURYHG·LWHPV·EXW·GRH
V·QRW·QHFHVVDULO
\·UHIOHFW·WKH·YLHZV·RI·WKH·&RQIHUHQFH· ··
```

Google Snippet

[PDF] Chapter 4 Judicial Federalism

famguardian.org/publishedauthors/Govt/JudConfOfUS/CH04.pdf ▾

The **Judicial** Conference of the United **States** has approved the recommendations and implementation strategies in this ... UDICIAL **federalism relies** on the prin ciple that the **state and federal** courts .... committed to the **principle of judicial** feder.

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OCR Parser to new TextRecognizer Component

```
{
  "engines": {
    "jina": {
      "openai-vlm-parser": {
        "baseUrl": "https://…",
        "model": "jina-ocr-v1",
        "apiKey": "${env:JINA_API_KEY}"
      }
    }
  },
  "text-recognizers": [
    {
      "engine": "jina",
      "checks": [
        { "min-image-size": {
          "minTotalPixels": 10000,
          "minShortSidePixels": 16 } }
      ]
    }
  ]
}
```

elastic

OpenAI API

- Some VLMs have been fine-tuned for OCR
- With Ollama and vLLM you can serve open-weight models with the OpenAI compatible API
- Or, open a ticket to wrap PaddleOCR like we do tesseract

Why modern OCR?

Credit: Scott Martens



https://www.elastic.co/search-labs/blog/ocr-model-jina-ocr-v1-document-parsing

Why modern OCR?

```
# PSA: your favourite author is back

New novels from our go-to writers are now hitting the shelves. But how do their second, fifth or 36th
book (we're looking at you Harlan Coben) compare to the classics? Team Cosmo finds out

COMPILED BY: LYDIA VENN

## The Midnight Feast by Lucy Foley
Out 6 June

The opening of a controversial wellness resort on the Dorset coast doesn't go to plan when secrets
start to emerge and a body is found. Told in non-chronological order, it follows the same formula as
The Guest List. But this story feels elevated beyond Foley's previous work. Alexandria Dale, fashion
writer

## Long Island by Colm Toibin
Out 23 May

This sequel revisits the characters from Brooklyn. When a man knocks on Ellis' door, it sets in motion
a series of events that bring her face-to-face with her past. Long Island is a relatable, witty and
hugely poignant look at what it means to be happy. Hannah Connolly, acting social media producer
```

https://www.elastic.co/search-labs/blog/ocr-model-jina-ocr-v1-document-parsing

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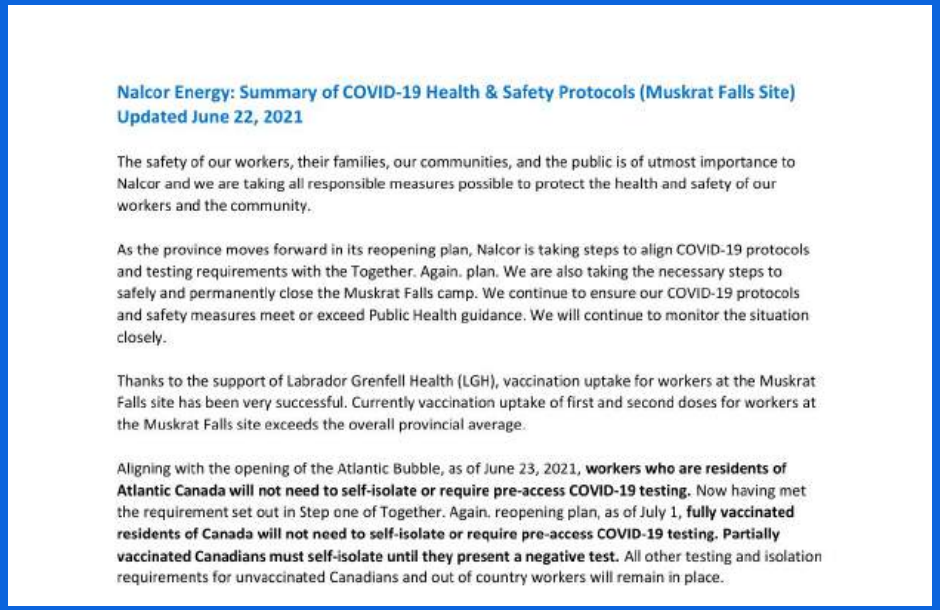
Why modern OCR?

- Headings
- Tables
- Math equations
- Better handling of multiple languages
- Handwriting

VLMs for OCR, a word of caution

Different models have different strengths

- Loops: 1. 2. 3. 1. 2. 3. 1. 2. 3.
- Fluent hallucinated text
- Some triggers:
  - Blank pages/non-text
  - Rotated Text
  - Big page/small font



Nalcor Energy: Summary of COVID-19 Health & Safety Protocols (Muskrat Falls Site)
Updated June 22, 2021

The safety of our workers, their families, our communities, and the public is of utmost importance to Nalcor and we are taking all responsible measures possible to protect the health and safety of our workers and the community.

As the province moves forward in its reopening plan, Nalcor is taking steps to align COVID-19 protocols and testing requirements with the Together. Again. plan. We are also taking the necessary steps to safely and permanently close the Muskrat Falls camp. We continue to ensure our COVID-19 protocols and safety measures meet or exceed Public Health guidance. We will continue to monitor the situation closely.

Thanks to the support of Labrador Grenfell Health (LGH), vaccination uptake for workers at the Muskrat Falls site has been very successful. Currently vaccination uptake of first and second doses for workers at the Muskrat Falls site exceeds the overall provincial average.

Aligning with the opening of the Atlantic Bubble, as of June 23, 2021, **workers who are residents of Atlantic Canada will not need to self-isolate or require pre-access COVID-19 testing.** Now having met the requirement set out in Step one of Together. Again. reopening plan, as of July 1, **fully vaccinated residents of Canada will not need to self-isolate or require pre-access COVID-19 testing. Partially vaccinated Canadians must self-isolate until they present a negative test.** All other testing and isolation requirements for unvaccinated Canadians and out of country workers will remain in place.

Rotate the image 225° and what?!?!?!
DeepSeek-OCR (October 2025)

```
1. A number of people are
waiting for the bus.
   - The bus is not here yet.
   - The bus is not here yet.
   - The bus is not here yet.
   - The bus is not here yet.
   - A number of people are
waiting for the bus.
   - The bus is not here yet.
   - The bus is not here yet.
   - The number of people is
increasing.
```

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Performance

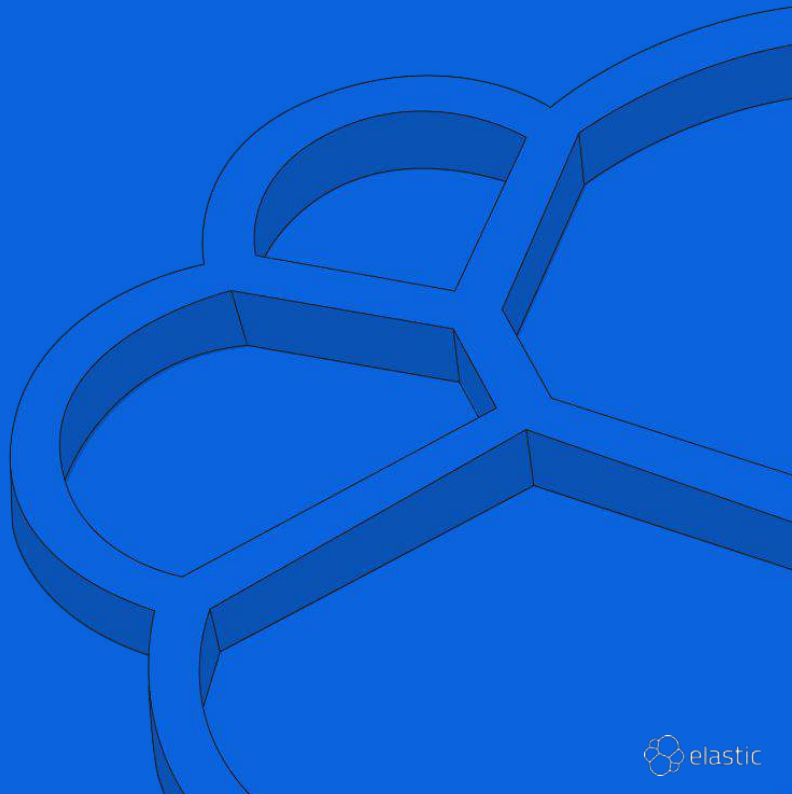
- Tesseract locally ~1 second per image
- Vlms range from 2 seconds - 30 seconds per image (ymmv). Possibly longer if hosting large model locally on a laptop (shout out to oar-ocr-vl)

Timeouts in 4.x

```
{
  "engines": {
    "jina": {
      "openai-vlm-parser": {
        "baseUrl": "https://example",
        "model": "jina-ocr-v1",
        "timeoutMillis": 180000}}},
  "text-recognizers": [
    { "engine": "jina" }],
  "parse-context": {
    "timeout-limits": {
      "totalTaskTimeoutMillis": 3600000,
      "progressTimeoutMillis": 120000
    }}}
```

elastic

Integrations for Multimodal Embeddings



Why Tika?!
We all know that embeddings are the step AFTER Tika!

- For text, could be
- Not so much for:
    - Page renderings
    - Embedded images and multimedia

Embedded file processing and multimedia chunking require parsing

Two options

- Inline during the parse
- Package all the embedded files in a Frictionless format zip for next pipeline stage

Option 1: Inline During the Parse



Chunking

- Text: user configured with breaks by length and/or document structure, headers, etc.
- PDF: per page, if embedding on page renderings
- Audio/Video: ffmpeg to the rescue. Default 25 second window with 5 second overlap.

Setting Up Inference Engines

```
{
  "engines": {
    "jina": {
      "openai-embedding-engine": {
        "baseUrl": "https://example…",
        "model": "jina-embeddings-v5-omni-small",
        "apiKey": "${env:JINA_API_KEY}",
        "maxBatchSize": 16 }}},
  "inference": [
    {
      "id": "inference-id",
      "engine": "jina",
      "input": "IMAGES|PAGES|TEXT|MEDIA",
       …
```

Output - Full doc


```
[
  {
    "Content-Type": "application/pdf",
    "pdf:encrypted": false,
    "xmpTPg:NPages": 2,
    "tk:chunks": [
      { … },
      { … },
    ],
    "tk:content": "Revenue grew 15%..."
  },
  {
    "Content-Type": "application/pdf",
    "tk:embedded-resource-path": "/embeddedPDF.pdf"
…
```

Chunk

```
{
  "text":       "Revenue grew 15%...",
  "vector":     "P4AAAEAAAABAQAAA…",
  "producer":   "text-vectors",
  "modality":   "text",
  "locators":   { … }
}
```

Locators

```
{“locators": {
  "text": [ { "start_offset": 1300,
    "end_offset": 2800 } ]
}}

{"locators": {
  "paginated": [ { "page": 2 } ]
}}

{"locators": {
  "embedded": [ { "id_path": "/1",
    "name": "image1.png" } ]
}}

{"correlator": "t:20000-45000",
"modality": "audio",
"locators": {
  "temporal": [ { "start_ms": 20000,
    "end_ms": 45000 } ]
}}
```

Option 2: All the embedded files in a zip for downstream processing and retrieval



PDF with Missing Unicode Mappings

Extracted text:

Constrained Least Squares Linear Spectral Unmixture by the Hybrid Steepest Descent Method

Nobuhiko Ogura* and Isao Yamada**

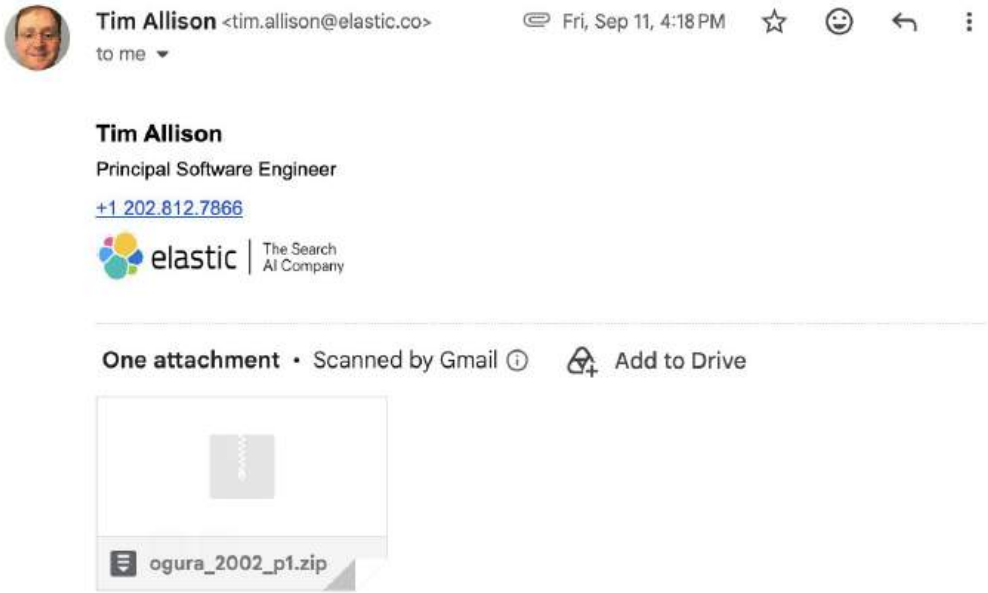
1 Introduction

A closed polyhedron is the intersection of finite number of closed half spaces, i.e., the set of points satisfying finite number of linear inequalities, and is widely used as a constraint in various application, for example specifications or constraints in signal processing or estimation problems, resource restrictions in financial applications and feasible sets of probability distributions. By the progress of the convex analysis and the fixed point theory of nonexpansive mapping, a number of convex projection based algorithms are proposed (for example, Bauschke et al, 1997; Combettes, 1993; Yamada et al, 1998–2002). In this paper, to apply efficiently such methods to problems with polyhedral constraints, we propose a simple solution to the problem of the best approximation to the certain polyhedron. By applying this solution to the the hybrid steepest descent method(Ogura et al, 2002; Yamada et al 1998; Yamada 2001), we also present two algorithms for a linear spectral unmixing problem. The proposed method enable us to deal with constraints and variety of cost function (for example, least square residual, Kullback-Leibler Divergence) as well as various a priori knowledge with great flexibility.

```
*,+-).'/01,23*.
457698;:;<>=75?&@78;ACBD(B7E;FHGJICBK5M
LNBKOPBKF;BDJD
QRS.TVU9WNXMY[Z\T]^W_S
`badc 5KICedFgfh5 cji ...
```

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Email with an attached zip containing a PDF




```
ogura.eml
├── image.png (signature icon)
└── ogura_2002_p1.zip
    └── ogura_2002_p1.pdf
```

Tika server /unpack/all

```
response.zip
├── datapackage.json
├── rmeta.json
└── unpacked/
    ├── 0.eml   the source email
    ├── 1.png   email's signature icon   /1
    ├── 2.zip   ogura_2002_p1.zip        /2
    ├── 3.pdf   ogura_2002_p1.pdf        /2/3
    └── 4.png   image-0.png (1×1)
/2/3/4
```

Frictionless datapackage.json


```
{
 "name" : "ogura.eml",
 "created" : "2026-10-14T07:53:56Z",
 "resources" : [ {
   "name" : "rmeta.json",
   "path" : "rmeta.json",
   "mediatype" : "application/json",
   "bytes" : 11025,
   "hash" : "sha256:36aace…"
   }, …, {
   "name" : "unpacked-1.png",
   "path" : "unpacked/1.png",
   "mediatype" : "image/png",
   "bytes" : 12086,
   "hash" : "sha256:8c3882…",
   "tk:embedded-id" : 1
   }, …,
```

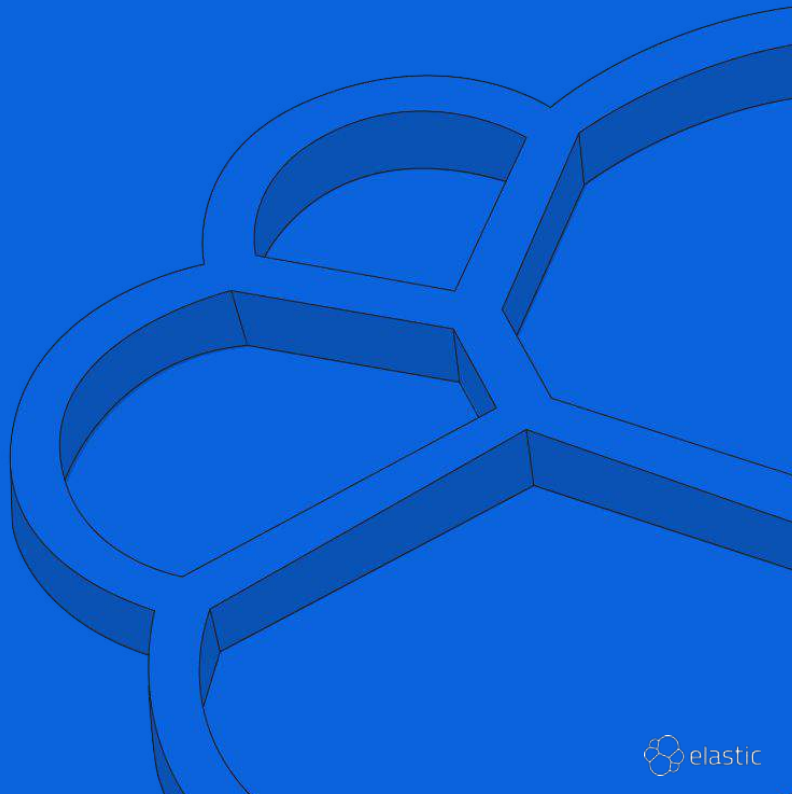
Snippets from rmeta.json

```
[
  { "Content-Type": "message/rfc822",
    "tk:resource-name": "Ogura 2002 p1.eml"…
  },
  { "Content-Type": "image/png",
    "tk:embedded-id": "1",
    "tk:embedded-id-path": "/1",
    "tk:embedded-resource-path":
      "/embedded-1.png" },
…zip file embedded-id=2
…pdf file embedded-id=3
  { "Content-Type": "image/png",
    "tk:resource-name": "image-0.png",
    "tk:embedded-id": "4",
    "tk:embedded-id-path": "/2/3/4",
    "tk:embedded-resource-path":
      "/…/image-0.png" },...
```

Snippet of PDF entry

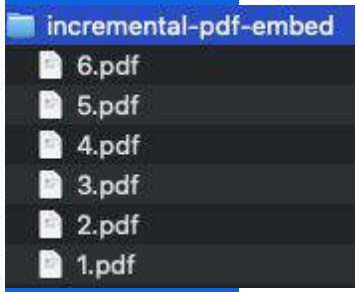

```
{
  "Content-Type": "application/pdf",
  "dcterms:created": "2002-11-01T13:24:21Z",
  "pdf:ocr-page-count": "1",
  "pdf:[...]percentage-unmapped-unicode-chars":
    "1.0",
  "pdf:producer": "GNU Ghostscript 5.10",
  "tk:embedded-id-path": "/2/3",
  "tk:embedded-resource-path":
    "/ogura_2002_p1.zip/ogura_2002_p1.pdf",
  "xmpTPg:NPages": "1",
  "zip:compressed-size": "43988",
  "tk:content": "Constrained Least Squares
Linear Spectral Unmixture by the Hybrid
Steepest Descent Method Nobuhiko Ogura’ and
Isao Yamada” 1 Introduction …"…}
```

Tika for agents: Treasure hunting



E-Discovery: “Claude, go find examples where a client might be surprised in discovery: incremental updates”





any such liability shall be limited to the

1.pdf: Edited June, 2004

aggregate amount of eur 1.000.000.

2.pdf: Edited March, 2005

aggregate amount of eur 100.000.

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“Claude, keep going…”

- Comment in a PDF: “did the redactions get applied or just marked”
- Username leakage in file paths: C:\Users\…\stuff\sector5.dwg
- Alaska Police Bulletin with embedded .doc of the original file… “Company: ConocoPhillips”
- Legislative working document with all details embedded as xml
- Tech tip 50301 → Tech tip 60601

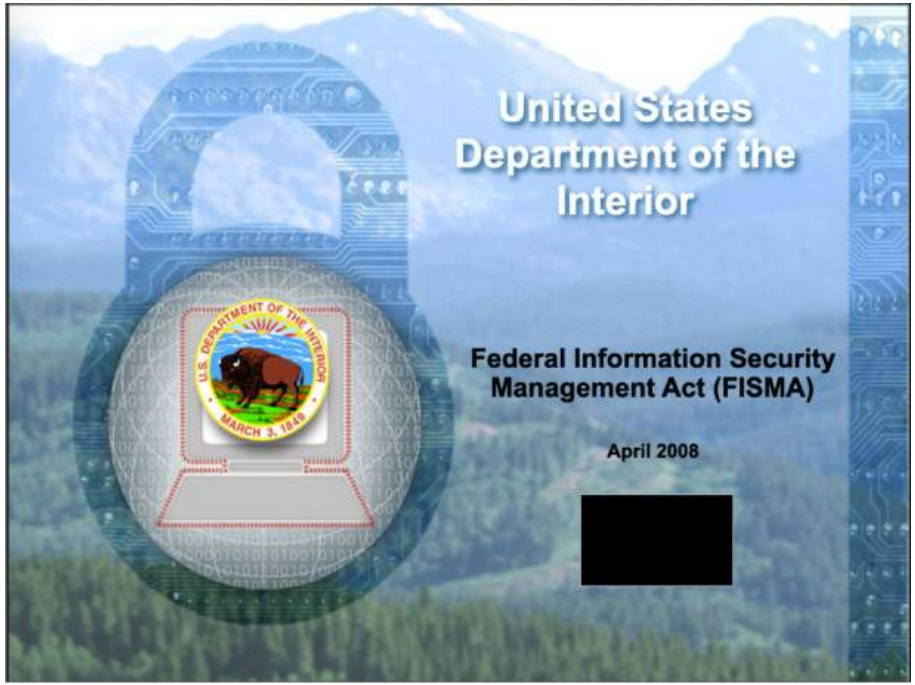
Bringing it all together: govdocs1 261779.ppt

Query: “no such agency”




```
{ "Content-Type": "image/png",
  "tiff:ImageLength": "72",
  "tiff:ImageWidth": "72",
  "tk:embedded-id": "121",
  "tk:embedded-id-path": "/20/121",
  "tk:embedded-resource-path":
    "/embedded-1.ppt/image-50.png",
  "tk:page:number": "94"}
```

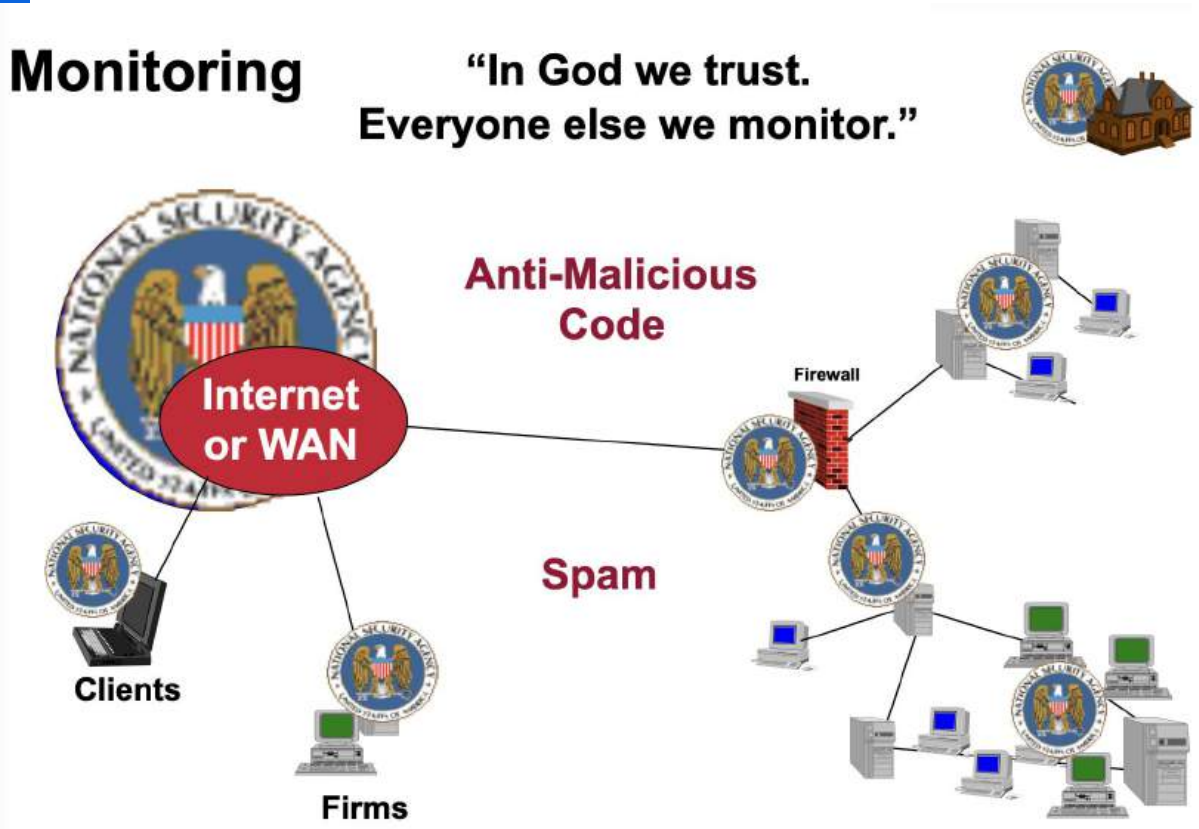
Bringing it all together, Container file




```
261779.ppt                    FISMA overview, 20 slides
└── embedded-1.ppt        138-slide deck     /20
    └── slide 94
        └── image-50.png        72×72        /20/121
```

elastic

Bringing it all together, contractor's deck, slide 94



elastic

New Tricks, Old Truths

If content extraction fails, you'll never know what you can't find



Many thanks!

- Tilman Hausherr
- Nicholas DiPiazza
- Julien Nioche
- Tika community

And of course, Elastic and my amazing colleagues!

TODOs

- Integrate vectors with other emitters, like Solr and OpenSearch
- Vespa emitter?
- ASR engines as text-recognizers
- Multivector chunks
- Solve PDFs

- What else?

Thank you!



Resources



Fetchers

- File system
- Http/https
- AWS S3
- Azure Blob Storage
- Google Cloud Storage
- Google Drive
- Microsoft Graph (SharePoint, OneDrive)
- Atlassian (Confluence/Jira)

Emitters

- File system
- AWS S3
- Azure Blob Storage
- Google Cloud Storage
- Elasticsearch
- Opensearch
- Apache Solr
- Jdbc
- Apache Kafka

Links

- Tika site
- Draft Apache Tika 4.2.0 site